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Role: Aspiring Data Analyst
Project Type: End-to-End Data Analytics Case Study
Tools Used: Python | MySQL | Power BI

Customer Shopping Behaviour Analysis

(End-to-End Data Analytics Project)

1. Project Overview

This project analyses customer shopping behaviour to identify purchasing patterns, revenue drivers, subscription impact, and product performance.
The project was executed using Python for data cleaning and transformation, MySQL for business query analysis, and Power BI for dashboard visualization.
The objective was to derive actionable business insights from raw customer transaction data.

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2. Tools & Technologies Used

- Python (Pandas, SQLAlchemy, PyMySQL)
- MySQL
- Power BI
- Jupyter Notebook
- CSV Dataset

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3. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features: -
  - Customer demographics
    (Age, Gender, Location, Subscription Status)
  - Purchase details
    (Item Purchased, Category, Purchase Amount, Season, Size, Colour)
  - Shopping behaviour
    (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

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4. Exploratory Data Analysis using Python– Data Cleaning

Data Exploration

- Checked dataset structure using df.info()
- Generated statistical summary using df.describe()
- Identified missing values using df.isnull().sum()

Handling Missing Values
Filled missing Review Rating values using median rating per category to maintain category-level consistency.

Column Standardization

- Converted column names to lowercase
- Replaced spaces with underscores for SQL compatibility

Age Group Segmentation:
Customers were segmented into:

- Young Adult
- Adult
- Middle Aged
- Senior

Using quartile-based binning.

Purchase Frequency Conversion:
Mapped textual purchase frequency into numerical days for analysis.

Data Cleaning:
Removed redundant column promo_code_used after verifying duplication with discount_applied.

Data Export to MySQL

- Established database connection using SQLAlchemy
- Uploaded cleaned dataset to MySQL
- Created table: customer

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5. MySQL – Business Problem Solving

Solved multiple real-world business questions using SQL, including aggregation, CASE statements, CTEs, and window functions.

Key Business Insights

1. Revenue comparison by gender
2. High-value customers using discounts
3. Top 5 highest-rated products
4. Shipping type comparison (Standard vs Express)
5. Subscription impact on revenue and average spend
6. Discount effectiveness by product
7. Customer segmentation (New, Returning, Loyal)
8. Top 3 products per category using CTE & ROW_NUMBER()
9. Repeat buyer subscription analysis
10. Revenue contribution by age group

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6. Power BI Dashboard Analysis

Created an interactive dashboard displaying:

KPI Metrics

- Total Customers: 4,000
- Average Purchase Amount: $59.76
- Average Review Rating: 3.75

Visualizations

- Total Revenue by Category
- Sales by Category
- Subscription Distribution
- Revenue by Age Group
- Sales by Age Group

Filters Used

- Gender
- Category
- Shipping Type
- Subscription Status

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7. Key Insights

- Clothing category generates the highest revenue
- Young Adults contribute the highest revenue
- 73% of customers are non-subscribers
- Not Subscribed customers show higher spending behaviour
- Clothing and Accessories drive the majority of sales

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8. Business Recommendations

- Target Young Adults with personalized campaigns
- Increase subscription adoption through loyalty benefits
- Promote high-rated products
- Optimize shipping strategies
- Focus marketing investment on top-performing categories

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9. Conclusion

This project demonstrates end-to-end data analytics capability, including data cleaning, feature engineering, database querying, visualization, and business insight generation.

It reflects a complete analytics workflow from raw data to executive-level dashboard reporting.

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10. Skills Demonstrated

- Data Wrangling using Pandas
- Feature Engineering
- SQL Aggregations
- CTE & Window Functions
- CASE Statements
- Data Visualization
- KPI Reporting
- Analytical Thinking